

The Comprehensive Evaluation Model for Earthquake Emergency Shelter

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ABSTRACT

The most important thing facing all levels of government after earthquake is to do everything possible to rescue injured people and evacuate those who are in the dangerous area to the safe places as soon as possible. What kinds of shelters are selected to be appropriate as earthquake shelters? This is an issue emergency management agencies have to deal with. For this reason it is necessary to set an evaluation model for earthquake emergency shelter (EES) so as to provide ground for decision making in the selection of earthquake emergency shelters. The objective of this study focuses on establishing a comprehensive evaluation model for the selection of earthquake emergency shelters. A total of three evaluation indicators are picked up: capacity of shelter, quality of facility, and accessibility. The definitions and quantitative calculation procedure of these indicators are elaborated. The weight of each indicator is estimated based on Analytic Hierarchy Process (AHP). A case study is also conducted to substantiate the validity and usefulness of the model. The result of this study can provide technical support in the selection of shelters for earthquake.

INTRODUCTION

Earthquake brings about more devastating damages for cities as compared to other disasters. Serious earthquakes usually cause family to lose their loved ones, huge economic losses, interruption of water supply, blackout, paralysis of transportation system, and chaos of social and economic life.

There are many seismic belts in China. The data from report show that there are 23 provincial capitals and 2/3 cities with population more than 1 million lie in 7-intensity earthquake belt (Lin, C.Y. 2010). Recent years,

earthquakes have already caused great damages to cities. For example, the earthquake in Wenchuan (Sichuan province of China) in 2008 claimed more than 80,000 lives and the economic loss was estimated over twenty billion US dollars.

After earthquake it is necessary to evacuate people who are in dangerous area to a relatively safe place with all kinds of necessary provisions, such as medical care facilities, lighting, public toilets, drinking water and food. This place is defined in this study as Earthquake Emergency Shelter (EES). EES is a place that can provide a safe place for the evacuees and meet the basic needs of life and emergency rescue after careful planning, construction and management (Qi, Y., 2005). Selection for EES is significant for emergency rescue and reconstruction after the shock (Xiu, J.G., 2001). However, how to choose the appropriate shelters after earthquake becomes an issue facing decision makers. It is crucial to conduct comprehensive evaluation for shelters so as to provide basis for decision making in the selection of EES. It will be beneficial for enhancing the planning and construction of EES and for improving the comprehensive emergency response ability of cities.

OBJECTIVES OF THE STUDY

Earthquake emergency shelter (EES) plays a very important role in earthquake emergency evacuation and rescue management. However, at present decision-making on the selection of earthquake shelters is usually arbitrary without appropriate methodology. The existing public parks and open spaces are generally chosen as earthquake shelters in the city of Beijing, without conducting technical analysis. To overcome this deficiency it is necessary to set principles of selection and build comprehensive evaluation model to minimize the arbitrariness.

It is evident that the selection of earthquake shelter is quite different from that of other disasters. Generally speaking earthquake shelter is featured with longer time of stay and daily necessity such as drinking water, food, medical care. As a result, the choice of emergency shelters should be made according to the particular requirement.

In this paper, a comprehensive evaluation model for the selection of EES is proposed. The objective of this study is to select appropriate and reasonable indicators for EES in the comprehensive manner. The model can be used for disaster officials to select suitable shelters for earthquake emergency purpose.

LITERATURE REVIEW

In her analysis of the critical affecting factors on disaster response ability of city, Wang Xia (Wang, X., 2009) divides the emergency process into three stages and proposes 23 assessment indexes to assess the disaster response ability

of cities. A comprehensive evaluation model over city's disaster response ability is built based on AHP.

Wu Zongzhi et al. conduct researches on evaluating emergency adaptation ability of shelter (Wu, Z.Z., 2005). He selects 18 assessment indexes from planning, internal hardware and outward software environment.

Bao Shengping carries out a study on adaptation assessment of urban refuge in Taiwan (Bao, S.P., 2005). He uses grey correlation analysis theory to do adaptation assessment for shelters and uses Jiayi city in Taiwan as an example.

Chen Sudan et al. analyze the security of shelters in Taiyuan city Shanxi province (Chen, S.D., 2006). In his opinion, there are two important safety factors that affect the safety and reliability of emergency evacuation and rescue significantly: geological factor and public facility factor.

Zhou Xiaomeng et al. study the layout optimization of shelter and propose network optimization model (Zhou, X.M., 2006). The principles of how to determine the number and capacity of shelters are given in this study.

Ye Mingwu et al. perform research on refuge accessibility of park in Shanghai based on GIS (Ye, M.W., 2008). Based on the theory of space accessibility and refuge request of disaster prevention, using ArcGIS technology, a quantitative study on the balance relationship between the existing service of emergency refuge of inner-city parks and refuge demand of residents in Shanghai is conducted.

METHODOLOGY

To evaluate EES in a comprehensive and relatively objective way it is crucial to pick up evaluation indicators among many available candidates. The selection of indicators is in accord to the following principles. First, the number of indicator should be as few as possible. Second, indicators should be independent with each other. Third, indicators should be quantifiable in estimation.

Based on the three principles and the specific demand of earthquake emergency three assessment indexes are selected. They are capacity of shelter, quality of facility and accessibility of shelter.

Interpretations of selected indicator

Capacity of shelter (B_1): Compared to other natural disasters, there will be much higher evacuation demand for earthquake emergency. As a safe refuge for people to get away from danger and wait for rescue, the EES must be large enough to accommodate as many people as possible. The larger the shelter is, the stronger emergency response ability is. Through literature review, it is found that capacity is an indispensable factor that must be considered in the study of

emergency shelter planning and deployment.

Quality of facility (B_2): Quality of facility is determined by three factors: provision of daily necessities, communication service and medical care facilities. The provision of daily necessities concerns with the supply of drinking water, food, tents and toilets. Communication service mainly encompasses deployment of guiding signs, emergency power supply facilities and communication facilities such as telephones and radios etc. With regard to the medical care the shelter should provide necessary medicines, equipments and facilities to conduct medical treatment and operation. It should be noted that daily necessity (C_2), communication service (C_3) and medical care facility (C_4) are difficult to quantitatively estimate. Questionnaire survey and scoring by professionals are combined to measure these indicators as substitutes.

Accessibility of shelter (B_3): Accessibility of shelter refers to the easiness for an emergency vehicle getting to the shelter from the affected area. The accessibility is associated with the distance from the affected place to shelter (C_5), number of connection of shelter to roadways (C_6) and category of highway connecting to shelter (C_7). Lessons have been learned from Wenchuan earthquake that due to the landslide blocking the roadway called “life-saving line” the rescue team couldn’t access ruined area to find survivors. This demonstrates how important of accessibility of shelter is in life-saving activity.

Definitions and calculations of the selected indicators

Capacity of shelter: Capacity is defined as the largest number of evacuees that an EES can accommodate. It is determined by the available refuge area of shelter and average refuge area per person. In order to ensure certain level of service quality, the average refuge area for each evacuee is taken as 3 m^2 . The formula for shelter capacity is given in equation 1. In the formula, the shelter’s available refuge area is 70% discounted, in order to leave some cushions to offset inaccuracy in number of evacuees in the prediction.

$$C = \frac{S \times 70\%}{3 \text{ m}^2} = \frac{(\text{available area} - \text{water area}) \times 70\%}{3 \text{ m}^2} \quad (1)$$

Where, C-Capacity of shelter;

S-Available refuge area (m^2). It is calculated by subtracting water area from the total available area.

Inside central city of Beijing, there are 30 shelters that were chosen as emergency shelters by 2010. In the study, we choose these 30 shelters to conduct capacity analysis. After data analysis, distribution of capacity of these shelters is shown in Figure 1. The 85th percentile and 50th percentile data are selected as threshold value and the capacity of shelter can be converted into standard score according to the threshold value. The score converting result is indicated in Table 1.

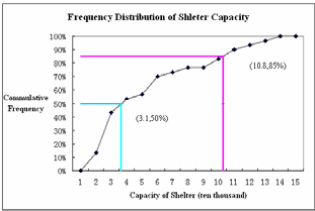


Figure 1. Frequency distribution for capacity of the 30 shelters.

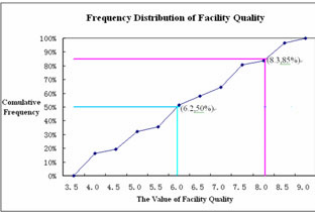


Figure 2. Frequency distribution for facility quality of 30 shelters.

Table 1. Standard Score for Capacity.

Capacity (C) (10^4)	≤ 3	(3,10)	≥ 10
Score of Capacity (C_s)	[0,50]	(50,80)	[80,100]

Quality of facility: In this study, quality of facility within EES is determined by provision of daily necessity, communication service, and medical care facility. Because it is hard to calculate these indexes quantitatively, professionals scoring method is used to measure these indexes. The evaluation results for quality of facility can be obtained by multiplying score of each sub-indicator given by each expert with corresponding weight. The calculation formula is shown in equation 2.

$$Q = x_2O_2 + x_3O_3 + x_4O_4 \quad (2)$$

Where, Q -Evaluation score for quality of facility of EES;

O_2, O_3, O_4 - Score given by expert for daily necessity, communication service and medical care facility of EES, $0 \leq O_i \leq 10$;

x_2, x_3, x_4 -Weight for daily necessity, communication service and medical care facility, $x_2+x_3+x_4=1$.

The evaluation value of quality can be converted into standard value. The threshold value for ranking is determined using the same method as applied for shelter capacity. The distribution of facility quality of these thirty shelters is given in Figure 2 and the conversion result is shown in Table 2

Table 2, Standard Score for Facility Quality.

Quality of Facility (<i>Q</i>)	≤6	(6,8)	≥8
Standard Score (<i>Q_s</i>)	[0,50]	(50,80)	[80,100]

Accessibility of shelter: Accessibility of shelter is inversely proportional to evacuation distance that is proportional to the number of accesses and category of connecting roads. The calculation formula for accessibility is shown in equation 3.

$$A = \frac{\sum_{i=1}^n k \times D_i}{l}$$

(3)

- Where, *A*-Accessibility of shelter;
n-Number of accesses of shelter;
l-Evacuation distance between affected place and shelter, (km);
D_i-Conversion value for road classification of road connecting to *ith* access;
k-Number of roads connecting to *ith* access.

The conversion value for road classification (*D*) referred in equation 3 is given in Table 3.

Table3. Conversion Value of Highway Category.

Highway Category	Freeway/ expressway	Major/ minor Road	Local street
Conversion Value <i>D</i>	3	2.5	2

After calculating accessibility of shelter, the value of accessibility can be converted into standard value. The threshold value for ranking is determined using the similar method as that for shelter capacity. The distribution of accessibility of these thirty shelters is given in Figure 3 and the conversion standard is shown in Table 4.

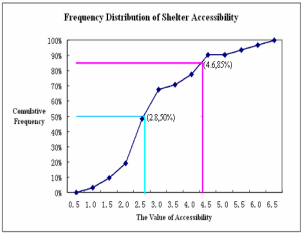


Figure 3. Frequency distribution for accessibility of 30 shelters.

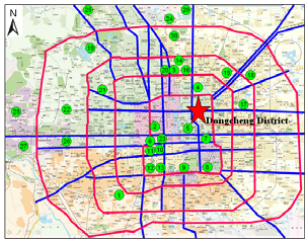


Figure 4. Location of the 30 shelters.

Table 4. Standard Score for Accessibility.

Accessibility (A)	≤ 2.8	(2.8,4.6)	≥ 4.6
Stand Score of Accessibility (C_s)	[0,50]	(50,80)	[80,100]

Weight calculation of indicator

Analytic Hierarchy Process (AHP) theory is used to calculate the weight for each indicator. The procedure for weight calculating is summarized as following.

First, assess the relative importance among different indexes that belong to the same criterion layer and then structure the comparison matrix. Then, based on the comparison matrix, calculate ranking weight for each index using square root method and conduct consistency test for comparison matrix.

In the study, in order to calculate the value of x_2, x_3, x_4 mentioned in equation 2 and the ranking weight of capacity (B_1), quality of facility (B_2) and accessibility (B_3), two comparison matrix are needed and they are structured by expert scoring method.

Thirty experts are invited in the study to structure the comparison matrix, including traffic engineers, safety engineers, road and bridge designers and transportation professors. Based on the comparison matrix given by different expert, the final comparison matrix is obtained using group decision method. On the basis of the final comparison matrix, the value of x_2, x_3, x_4 and ranking weight p_1, p_2, p_3 for B_1, B_2 and B_3 are calculated as well as consistency test for comparison matrix is conducted. The final calculation results are

$$X = (x_2, x_3, x_4) = (0.598, 0.174, 0.228)$$

$$P = (p_1, p_2, p_3) = (0.500, 0.235, 0.265)$$

Model build-up

Based on the special demand of earthquake emergency, three assessment indicators are selected and they are calculated quantitatively. The ranking weight for each indicator is calculated using AHP theory. On the basis of all these analyses, the comprehensive evaluation model for EES is established using linear weighted theory. The model is shown in equation 4.

$$Z = p_1 C_s + p_2 Q_s + p_3 A_s \tag{4}$$

Where, Z —Comprehensive score of shelter, $0 \leq Z \leq 100$;

C_s —Standard score of capacity of shelter, $0 \leq C_s \leq 100$;

Q_s —Standard score of quality of facility of shelter, $0 \leq Q_s \leq 100$

A_s —Standard score of accessibility of shelter, $0 \leq A_s \leq 100$.

p_1, p_2, p_3 —The ranking weight for capacity of shelter, quality of facility, accessibility of shelter, $p_1 + p_2 + p_3 = 1$.

Based on the calculated score, relatively objective evaluation upon comprehensive response ability of shelter can be carried out. Emergency officials can select EES according to the standard and advice designed in Table 5. If the evaluation score is between 80 and 100, it means the comprehensive emergency response ability of shelter is excellent and the shelter can be selected. If the score is between 60 and 80, it means the comprehensive emergency response ability of shelter is good and the shelter can be used for earthquake emergency. When the score is between 40 and 60, it means the shelter is not very suitable for earthquake emergency; but if shelters with score higher than 60 can not meet the evacuation demand, the shelter can also be selected. If the score is lower than 40, the shelter can not be selected as EES. In a word, in the selection, shelter with higher comprehensive score has higher likelihood to be chosen.

Table 5 Level of service of EES

Condition Level	Evaluation Score	Comments for selection
A	(80,100]	Preferred
B	(60,80]	Good
C	(40,60]	Pending
D	≤ 40	Reject

CASE STUDY

In the study, thirty shelters mentioned above are taken as examples to apply the comprehensive evaluation model. Distribution of the thirty shelters inside Beijing is shown in Figure 4. Field survey for these shelters was conducted. Because of the limit of paper, all survey data are not listed.

According to the survey data, we calculate capacity and accessibility of each shelter using equation 1 and 3. At the same time, thirty professionals are invited to evaluate the quality of facility. Then we convert the value of capacity, accessibility and quality of facility into standard value according to the threshold value proposed in the study. Finally, we calculate the comprehensive evaluation value of each shelter using equation 4. Shelter with the high score should be considered first during decision process for EES choice.

It can be seen from Table 6 that the emergency response ability of the first three shelters is excellent and they can be selected for EES. Emergency response ability of shelters from the 4th to 13th is good and these shelters can be selected as EES according to the specific evacuation demand. Shelters from 14th to 23th are not very suitable for earthquake emergency, and are staying at the pending status. The remaining shelters can not be chosen as EES because the score is lower than 40.

Table 6. The Comprehensive Evaluation Result for These Thirty Shelters.

Ranking	NO.	Name	C	C _s	Q	Q _s	A	A _s	Z
1	21	D.P. Park	9	84	8.58	82	2.02	86	84.06
2	14	Y.D.D. Park	15.7	100	5.92	50	4.31	76	81.89
3	3	M.D.R. Garden	13.3	92	6.46	60	4.62	81	81.57
4	2	S.C. Open Space	12.3	88	3.61	36	6.21	95	77.64
5	7	M.C.Q. Park	6	70	7.47	73	4.50	83	74.15
6	5	H.C. Park	4.5	65	6.79	65	5.83	92	72.16
7	17	C.Y. Park	11.5	86	8.01	80	2.09	38	71.87
8	23	C.C. Gym	5	60	8.45	81	3.64	75	68.91
9	26	G.J.S. Garden	10.6	84	4.89	45	2.90	55	67.15
10	19	H.D. Park	9.3	78	6.62	60	2.24	40	63.70
11	27	B.H. Square	10	80	4.59	43	2.50	47	62.56
12	15	T.Y.G. Park	8.6	76	5.74	48	2.76	50	62.53
13	1	F.Y. Park	4.7	65	7.09	67	2.25	46	60.44
14	25	W.Q. Park	2.5	40	8.29	81	1.12	66	56.53
15	24	D.S. Park	4	45	7.61	75	2.05	59	55.76
16	20	M.D. Park	4.3	58	7.08	67	2.19	40	55.35
17	4	D.T. Park	2.7	50	7.08	69	2.24	45	53.14
18	18	J.T. Open Space	6.4	70	3.61	34	1.63	30	50.94
19	12	F.Y. Park	2.2	35	6.35	56	4.38	76	50.80
20	13	W.S. Park	2.1	35	6.25	52	4.33	76	49.86
21	30	Olympic Park	2	32	8.09	82	2.76	50	48.52
22	29	Y'an Park	2.8	48	4.88	44	2.44	43	45.74
23	22	Y.G. Park	2.6	45	7.43	73	1.07	20	44.96

Note: In the study, Dongcheng District (marked in Figure 4) is the assumed evacuation area to calculate accessibility of the thirty shelters.

CONCLUSION AND FURTHER STUDY

The major work carried out in the study is summarized as following.

- (1) Three assessment indicators are selected. Weight for each indicator is calculated based on the Analysis Hierarchy Process (AHP)
- (2).The comprehensive evaluation model is built up using linear

weighted theory. Beside, a recommended table of level of service of EES is given.

(3)Thirty existing shelters that have already been constructed as emergency shelters in Beijing are selected as samples to apply and test the comprehensive evaluation model.

Compared to the previous studies, the selected assessment indicators are more objective and detailed. The specific demand of earthquake emergency and effect caused by traffic on emergency response ability of shelters are considered during indicator selection. A recommended table of level of service of EES is given, which can guide the emergency officials to select the suitable shelters for earthquake emergency. The evaluation result can provide technical support for emergency officials to select suitable shelters for earthquake emergency evacuation and rescue as well as reduce the arbitrary determination and subjectivity during the decision-making process. It will improve the existing shelters and guide the construction and planning of shelters in the future.

However, it is a preliminary study. Although some positive results have been obtained from this study, there is still room for improvement. In order to determine the ranking weight of each indicator, thirty experts including traffic engineers, safety engineers, road and bridge designers and transportation professors are invited to structure the comparison matrix. The number of experts in the study is not adequate. It is suggested that more experts be invited so as to establish more scientific and reasonable evaluation model.

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